Sushant S. Mahajan

Solar Astrophysicist • Data Scientist • AI for Science

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SUMMARY

Research scientist and computational physicist building data-driven systems for solar physics, scientific inference, and time-series analysis.

EXPERIENCE

Research Scientist, W. W. Hansen Experimental Physics Laboratory, Stanford University 2024–present

- Lead research on solar dynamics and large-scale flows from HMI/SDO observations.
- Produce new metadata and analysis workflows for solar magnetogram and velocity-field data.
- Contribute to an agentic AI-powered JSOC data-commons effort for public query and visualization.

Postdoctoral Fellow, W. W. Hansen Experimental Physics Laboratory, Stanford University 2022–2024

- Published first-author work on differential rotation, torsional oscillation, and meridional flow.
- Helped deliver the Stanford Solar Observatories Group data-sharing effort recognized in 2026.

Solar Physics Postdoctoral Fellow, Institute for Astronomy, University of Hawai'i 2019–2022

- Built analysis pipelines for active-region inflows, solar-cycle trends, and helioseismic flow inference.

SELECTED IMPACT

- NASA Silver Group Achievement Award for JSOC flood recovery efforts 2025
- Stanford University Libraries Data Sharing Prize, Stanford Solar Observatories Group 2026
- Editor's Choice, *Solar Physics*, for *The Sun's Large-Scale Flows I* 2024
- Prepared and published the SWAN-SF benchmark dataset used in solar-flare AI work 2020
- Presentation awards: Best Young Presenter Award (IAU Symposium 340), Best Young Scientist Poster Award (IAU Symposium 335), Honorary Mention (SPD meeting) 2018–2016

SELECTED TALKS & MEDIA

"Unveiling the Sun: Exploring the Wonders of Solar Astrophysics," KIPAC public lecture and "The Influence of Active Regions on Plasma Flows Inside the Sun," UCAR/HAO colloquium 2023

"Unveiling the Solar Poles: Addressing Projection Artifacts and Revealing Circumpolar Features in HMI Magnetograms," AAS/SPD 246, Anchorage, AK 2025

KRON4, NBC Bay Area, ABC7 Bay Area, and *San Francisco Chronicle* coverage of the 2024 Northern Lights 2024

SKILLS

Python, MATLAB, Fortran, bash, OpenMP, MPI, C++, CUDA, scientific computing, time-series analysis, machine learning workflows, data analysis, visualization, and technical writing.

EDUCATION

Ph.D. in Astronomy, Georgia State University 2019
M.S. in Physics, Georgia State University 2017
M.Tech. in Engineering Physics, IIT (BHU) 2014